

Printers & Displays

Lesson 15 — Mechanisms and Display Terminology

JKSSB Computer Awareness

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Printers: Hard-Copy Output

- ▶ A **printer** is an output device that produces **hard copy** — output on a physical medium, usually paper.
- ▶ Output shown on a monitor is called **soft copy**; a printer turns it into permanent hard copy.
- ▶ Printers are compared in three main ways: by **mechanism** (dot matrix, inkjet, laser, thermal, 3D), by **impact or non-impact**, and by the **unit printed** (character, line, page).
- ▶ Choosing the right labels is the whole exam skill here.

Definition

A printer accepts output data from the computer and records it as a permanent, human-readable copy on paper or another medium.

Classifying Printers

Basis	Types
By mechanism	Dot matrix, inkjet, laser, thermal, 3D printer
By contact	Impact (strikes the paper) vs non-impact (no contact)
By unit printed	Character printer, line printer, page printer

The same printer carries several labels: a laser printer is non-impact, a page printer, and uses a laser mechanism.

Impact vs Non-Impact

Impact printer

- ▶ Forms characters by **striking** an inked ribbon against the paper
- ▶ Noisy; can print carbon copies
- ▶ Dot matrix, daisy wheel, drum/line printer

Non-impact printer

- ▶ Prints **without striking** the paper
- ▶ Quieter; better quality; no carbon copies
- ▶ Inkjet, laser, thermal

Trap alert

Impact is about **striking the paper**, not about how the image is formed. Dot matrix strikes the paper; inkjet, laser, and thermal do not.

Impact Printers

- ▶ An impact printer forms a character by **physically hitting** an inked ribbon against the paper.
- ▶ **Dot matrix**: pins hit the ribbon to build characters from dots.
- ▶ **Daisy wheel**: a spinning wheel of type characters, like a typewriter; good text, no graphics.
- ▶ **Line printers** such as the drum printer and chain printer strike a whole line at a time at high speed.
- ▶ Because it presses through, an impact printer can make **multiple carbon copies** of multi-part forms.

Dot Matrix Printer

- ▶ It is an **impact** printer. The **print head** contains rows of tiny pins.
- ▶ The pins strike an **inked ribbon** to form each character as a pattern of dots in a matrix.
- ▶ It is **noisy**, cheap to run, and can print **carbon copies** and multi-part invoices.
- ▶ Print quality is low and graphics are poor; still used for receipts and billing.

Trap alert

A dot matrix printer is an **impact** printer (it strikes the ribbon). Do not group it with inkjet, laser, or thermal.

Line Printers

- ▶ A **line printer** prints an entire **line** of text in one operation, not one character at a time.
- ▶ **Drum printer**: a rotating drum carries the character set; a hammer strikes the paper where each character is needed.
- ▶ **Chain (band) printer**: a moving chain or band of type characters runs past the hammers.
- ▶ They are high-speed **impact** printers used with older mainframes and large batch jobs.

Print units

Character printer → line printer → page printer, from smallest to largest unit printed.

Inkjet Printer

- ▶ A **non-impact** printer: it does not touch the paper.
- ▶ The print head sprays tiny **droplets of liquid ink** through fine nozzles onto the page.
- ▶ It is quieter than an impact printer and prints good-quality **full colour** images on many paper types.
- ▶ It is slower per page than a laser printer, and wet ink can **smudge**; ink cartridges need replacing.

Remember

Inkjet = **liquid ink** sprayed through nozzles; no toner, no striking.

Laser Printer

- ▶ A **non-impact** printer that uses a **laser beam** and **toner** (fine powder) in a xerographic process.
- ▶ The laser draws an image on a charged drum; toner sticks to it and is then fused onto the paper with heat and pressure.
- ▶ It is **fast** with sharp, high-quality text, and it is a **page printer** — it prints a whole page at a time.
- ▶ Toner is used instead of ink, so output does not smudge when wet.

Trap alert

Laser = **toner** + laser beam. Inkjet = **liquid ink**. Do not swap the two consumables.

Thermal Printer

- ▶ A **non-impact** printer that produces an image using **heat**.
- ▶ It heats specially treated **thermal paper** (or a heat-sensitive ribbon) to form characters.
- ▶ It is **silent** in operation and needs no ink or toner cartridge in the simplest design.
- ▶ Used for POS receipts, tickets, barcode labels, and older fax machines; usually limited to one colour.

Keep it non-impact

Thermal, laser, and inkjet are **non-impact**. Dot matrix is **impact**.

3D Printer

- ▶ A **3D printer** builds a solid three-dimensional object **layer by layer** from a digital model — this is **additive manufacturing**.
- ▶ It does not print ink on paper; it adds material (plastic, resin, or metal) until the object is formed.
- ▶ Used for prototypes, models, spare parts, and medical implants.
- ▶ The word “printer” is used, but the output is a physical object, not a document.

Plotters

- ▶ A **plotter** is an output device that **draws** line graphics, usually with one or more pens moving over the paper.
- ▶ It works with **vector** drawings, so it is ideal for large **engineering drawings**, maps, and CAD plans.
- ▶ Plotters handle big sheet sizes that ordinary printers cannot.
- ▶ A plotter draws lines; a printer lays down dots or characters. Both are output devices.

Trap alert

A plotter is an **output** device, not an input device. Do not confuse it with a scanner or a digitizer (TRAP-12).

Inkjet vs Laser (Near-Neighbour)

Point	Inkjet	Laser
Contact	Non-impact	Non-impact
Image formed by	Liquid ink droplets	Laser beam + toner
Unit printed	Character/line	Page printer
Speed	Slower	Faster
Strengths	Colour photos	Sharp text, offices

Both are non-impact. The usual trap swaps their consumables and speeds.

Impact or Non-Impact? Sort Each Printer

- ▶ **Impact:** dot matrix, daisy wheel, drum printer, chain printer.
- ▶ **Non-impact:** inkjet, laser, thermal.
- ▶ A **page printer** (such as a laser printer) prints a whole page at a time; a **line printer** prints a whole line at a time.
- ▶ 3D printers and plotters are in their own categories; they are not ordinary paper printers.

Trap alert

Only devices that **strike an inked ribbon** are impact. Inkjet, laser, and thermal never touch the paper.

Display Terminology: Pixel and Resolution

- ▶ A **pixel** (picture element) is the **smallest addressable element** of a display; every image is built from pixels.
- ▶ **Resolution** is the number of pixels the display can show, written as columns $\times$ rows (for example 1920×1080).
- ▶ **Higher resolution** means more pixels and a **sharper** image.
- ▶ Printer sharpness is measured in **DPI = Dots Per Inch**.

One idea

Pixels are the units; resolution is how many of them fill the screen.

Display Terminology: Refresh Rate and Response Time

- ▶ **Refresh rate** is how many times per second the display redraws the image; measured in **hertz (Hz)**.
- ▶ A **higher refresh rate** (for example 120 Hz) gives smoother motion and less flicker.
- ▶ **Response time** is how long a pixel takes to change from one state to another; measured in **milliseconds (ms)**.
- ▶ A **lower response time** means less motion blur or ghosting.

Trap alert

Refresh rate is in **Hz** (higher is better); response time is in **ms** (lower is better). Do not swap the units.

Brightness and Contrast

- ▶ **Brightness** is the amount of light a display emits; it is commonly measured in **nits** (candela per square metre, cd/m^2).
- ▶ **Contrast** is the difference between the brightest white and the darkest black a display can produce.
- ▶ The **contrast ratio** compares the two (for example 1000:1); a higher ratio gives deeper blacks and more vivid images.
- ▶ Brightness is how much light; contrast is the range between light and dark.

Summary of units

Resolution $\rightarrow$ pixels/DPI · refresh rate $\rightarrow$ Hz · response time $\rightarrow$ ms · brightness $\rightarrow$ nits.

Lesson 15: Recall

- ▶ A printer produces **hard copy**; monitor output is soft copy.
- ▶ **Impact** printers (dot matrix, daisy wheel, drum/chain line printers) strike an inked ribbon; **non-impact** printers (inkjet, laser, thermal) do not.
- ▶ Dot matrix = impact, pins + ribbon, carbon copies; inkjet = liquid ink sprayed; laser = laser beam + toner, a page printer; thermal = heat-sensitive paper.
- ▶ A 3D printer adds material layer by layer; a plotter draws vector line graphics for engineering drawings.
- ▶ Display terms: **pixel** (smallest element), **resolution** (pixels shown), **refresh rate** (Hz), **response time** (ms), **brightness** (nits), **contrast** (bright/dark ratio).